

NEO Mathematics Supporting Adult Guidance

Year 9 Unit: Pythagoras and the Hidden Length

Unit Driving Question: How can a right-angled triangle reveal a length we cannot measure directly?

Purpose of this guide

This guide is for teachers, Practitioners, parents, carers and other Supporting Adults. It explains the 15 session opportunities, the optional Reconnection Routes and the facilitation approach. It is not a script and does not assume the adult is a mathematics specialist.

Core mathematical story

The unit is organised around hidden measurement and area relationships: right triangle -> areas of squares -> relationship -> symbolic equation -> hidden lengths -> theorem selection -> embedded triangles -> indirect measurement.

The most important facilitation move

Notice first. Ask second. Explain only as much as the learner needs to keep thinking.

Using the route

The 15 sessions are opportunities, not a rigid conveyor belt. Sessions 3 and 8 are targeted Reconnection Routes. A learner who is secure with squares/roots or right-angle recognition may move past those sessions. A learner may also return to them later.

Session route at a glance

Seq	Session	Type	Key adult prompt
1	The hidden length	Core	Right angle first. Opposite side second. Name the hypotenuse third.
2	Squares on the sides	Core	Do not memorise the formula yet. Look at the three areas and ask what combines.
3	Reconnection: squares, roots and number structure	Reconnection	Use nearby perfect squares to estimate. Exact memory is helpful, but structure is more important.
4	From picture to equation	Core	Diagram → name the hypotenuse → write the square relationship. One step at a time.
5	Finding the longest side	Core	Square. Add. Root. Then check the answer is longer than each shorter side.
6	When the unknown is not the longest side	Core	Find c first. Ask which square area is missing. Subtract before the root.
7	Does the answer make sense?	Core	Before calculating, compare the missing side with the known sides.
8	Reconnection: angles and right triangles	Reconnection	Known angles first. Total 180° . Find the missing angle. Then decide whether Pythagoras is available.

Seq	Session	Type	Key adult prompt
9	Is Pythagoras allowed here?	Core	Ask one gatekeeping question: where is the right angle?
10	Measuring what is hard to reach	Core	Draw the triangle first. Label known lengths and units. Calculate only after the model is clear.
11	One diagonal, two routes	Core	You do not need to imitate the fastest-looking method. Choose a route you can defend.
12	A diagonal inside a shape	Core	Find the right angle. Trace two perpendicular sides. Join their endpoints.
13	From two dimensions towards three	Core	Triangle one first. Save the intermediate length. Triangle two second.
14	What can I now see that was hidden before?	Consolidation / Mastery	You do not need to prove everything in one sitting. Choose one strong piece of evidence and one next step.
15	How far can indirect measurement take me?	Extension / Portfolio	Choose one measurement question. Build one triangle. Test one route. Add limitations after the mathematics works.

Reconnection Routes

Squares, roots and number structure: use when square numbers, square roots or estimation are materially blocking the learner from interpreting or calculating a hidden length. Minimum return evidence: the learner can use enough square/root relationships to resume the Pythagoras route.

Angles and right triangles: use when the learner cannot reliably identify or justify a right angle. Minimum return evidence: the learner can use angle facts sufficiently to recognise a right triangle and decide whether Pythagoras is available.

Reconnection is not 'going backwards'. Name the route explicitly, keep the return point visible and return as soon as the prerequisite connection is usable.

Common misconceptions and supportive prompts

The hypotenuse is always the bottom side. Prompt: 'Where is the 90° angle? Which side is directly opposite it?'

Add lengths rather than square areas. Prompt: 'What do the squares on the sides represent? Are we combining lengths or areas first?'

Always add when using Pythagoras. Prompt: 'Is the missing square part of the total or is c^2 the total?'

A calculator answer must be correct. Prompt: 'Could that side be longer than the hypotenuse? What should the missing length be bigger or smaller than?'

Pythagoras works on any triangle. Prompt: 'Where is the proven right angle?'

Using the Pythagoras Explorer

The session pages use different presets and interactions rather than one repeated exercise. Invite the learner to predict on the Scratchpad before revealing or checking where possible. The aim is Predict -> Sketch -> Test -> Notice -> Revise.

Using the Scratchpad

The embedded NEO Maths Scratchpad is optional. It supports pen colours, width, eraser, undo/redo, Sticky Notes and mathematical backgrounds. Encourage colour coding where it reduces load: for example, one colour for the right angle, one for the hypotenuse, or three colours for square -> add/subtract -> root.

Regulation and access

If the learner does not want to speak: accept clicking, sorting, drawing, typed Sticky Notes or a saved Scratchpad image.

If visual load is high: close other panels, use the Rest routine, highlight only the required triangle or work on paper.

If calculation is blocking the concept: allow calculator use after the mathematical route has been chosen.

If the learner becomes worried about mistakes: use 'Let's test that idea' and make Undo/Redo visible. Mistakes are useful evidence about the relationship that needs attention.

Evidence and Mastery

Evidence may include an annotated triangle, Scratchpad image, colour-coded calculation, sorted theorem-selection cards, contextual model, verbal/audio explanation, repaired misconception or portfolio investigation.

Emerging: identifies some right-triangle features and follows a Pythagorean route with support.

Developing: uses the theorem in familiar cases and can find a hypotenuse or shorter side with prompts.

Secure: selects Pythagoras appropriately, calculates accurately and checks reasonableness.

Mastering: reveals embedded right triangles, compares routes, builds indirect-measurement models and evaluates assumptions or limitations.

Final portfolio piece

The final Indirect Measurement Portfolio Piece should contain a measurement question, a right-triangle model, known values and units, a Pythagorean calculation, a reasonableness check, an assumption and a limitation. The learner may present this through annotated images, typed text, audio-supported explanation or a mixed portfolio format.